Reverse Engineering Papyrus Segmentation Pipeline

Herculaneum Scroll CT Data

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Overview

This project implements a full processing pipeline for CT scan data of Herculaneum papyrus. The workflow includes dataset exploration, volume visualization, clustering analysis, and autoencoder-based reconstruction using Python scripts.

The system is designed for lightweight execution and patch-based learning from high-resolution volumetric CT data.

Pipeline

```
CT Volume      Normalization      Patch Extraction      Clustering
    Autoencoder Training      Reconstruction      Evaluation
```

Project Structure

```
project/
          dataset_exploration.py
          volume_visualization.py
          segmentation_pipeline.py
          experiment_pipeline.py

          data/


          figures/
                clustering_result.png
                reconstruction_*.png
                training_curve.png

          requirements.txt
```

Installation

1. Create Virtual Environment

```
python -m venv venv
source venv/bin/activate   % Linux / Mac
venv\Scripts\activate      % Windows
```

2. Install Dependencies

```
pip install -r requirements.txt
```

Dataset Setup

Download and extract the Herculaneum CT dataset and update file paths inside the scripts:

```
dataset_path = "path/to/Scroll1"
volume_folder = "path/to/volumes/"
data_path = "path/to/patches.npy"
```

Important: All paths are hardcoded and must be adjusted before execution.

Execution Order

Run the Python scripts in the following sequence:

1. Dataset Exploration

```
python dataset_exploration.py
```

Output:

- patches.npy

2. Volume Visualization and Patch Extraction

```
python volume_visualization.py
```

Output:

- patches$_s$*mall.npy*
- slice images

3. Segmentation Pipeline (Clustering)

```
python segmentation_pipeline.py
```

Output:

- clustering_result.png

4. Autoencoder Experiment

```
python experiment_pipeline.py
```

Output:

- reconstruction images
- training_curve.png

Model Summary

- Model: Convolutional Autoencoder
- Input: 128×128 grayscale patches
- Parameters: ∼4800
- Loss Function: Mean Squared Error (MSE)
- Optimizer: Adam
- Training: 3 epochs, batch size 8

Key Design Decisions

- Patch-based processing for memory efficiency
- Lightweight architecture for HPC constraints
- Unsupervised learning (no labels required)
- K-Means clustering for data augmentation

Limitations

- No ink or text detection module
- Limited dataset subset for training
- Loss of fine details due to compression

Future Work

- Integrate ink detection models
- Use perceptual or contrastive loss
- Train on full dataset

Conclusion

This project demonstrates that lightweight convolutional autoencoders can effectively learn structural representations from CT scans of papyrus, enabling meaningful reconstruction and analysis under computational constraints.